# Redwood Materials launches energy storage business and its first target is AI data centers

Tucked between two massive buildings in the hills of the Nevada desert, 805 retired EV batteries lie in neat formation, each one wrapped in nondescript white tarps — and hiding in plain sight.

A passerby might not realize this unassuming array is the largest microgrid in North America, that it's powering a 2,000 GPU modular data center for AI infrastructure company Crusoe, or that it represents the next big act of JB Straubel, the co-founder and CEO of Redwood Materials.

Redwood Materials announced Thursday during an event at its Sparks, Nevada, facility that it was launching an energy storage business that will leverage the thousands of EV batteries it has collected as part of its battery-recycling business to provide power to companies. And they're starting with AI data centers.

The new business, called Redwood Energy, is kicking off with partner Crusoe, a startup Straubel invested in 2021. The old EVs, which are not yet ready for recycling, store energy generated from an adjacent solar array. The system, which generates 12 MW of power and has 63 MWh of capacity, sends power to a modular data center built by Crusoe, a company best known for its large-scale data center campus in Abilene, Texas — the initial site of the Stargate project.

**Image Credits:**redwood materials

The scale of Redwood's battery-collection operation is staggering — and an opportunity. Redwood said it recovers more than 70% of all used or discarded battery packs in North America. Today, it processes more than 20 GWh of batteries annually — the equivalent of 250,000 EVs.

It has apparently been stockpiling batteries that aren't ready for recycling, with more than 1 gigawatt-hour worth in its inventory already. In the coming months, it expects to receive another 4 gigawatt-hours.

By 2028, the company said it plans to deploy 20 gigawatt-hours of grid-scale storage, placing it on track to become the largest repurposer of used EV battery packs.

Straubel's confidence in the endeavor was apparent in every detail of the launch event. To illustrate the commitment of Redwood — and by extension, Straubel — everything at the event, from the music and projection on the big screen to a laser light show that included giant Pac-Man ghosts navigating the rows of EV batteries, was powered by the microgrid.

**Image Credits:**kirsten korosec

"We wanted to go all in," Straubel said, breaking into a wide, toothy smile. Splashy effects for the event aside, the microgrid setup with Crusoe is not a demonstration project. Straubel said this is a revenue-generating operation, which was constructed in four months, and one that is profitable. He added that even more of these will be deployed with other customers this year.

"I think this has the potential to grow faster than the core recycling business," he said.

Redwood Materials has been on an expansion tear in recent years. The company, which has raised $2 billion in private funds, was founded in 2017 by Straubel, the former Tesla CTO and current board member, to create a circular supply chain.

The company started by recycling scrap from battery cell production and from consumer electronics like cell phone batteries and laptop computers. After processing these discarded goods and extracting materials like cobalt, nickel, and lithium that are typically mined, Redwood supplies those back to Panasonic and other customers. Over time, the company has expanded beyond recycling and into cathode production. Redwood generated $200 million in revenue in 2024, much of which comes from the sale of battery materials like cathodes.

The company's footprint has grown, too, and well beyond its Carson City, Nevada, headquarters. It has locked up deals with Toyota, Panasonic, and GM; started construction on a South Carolina factory; and made an acquisition in Europe.

Redwood Energy is the next step — and one that isn't tied to setting up its systems to be off-grid. The retired EV batteries can be powered by wind and solar, or they can be tied to the grid. In the case of the Crusoe project, the system is powered by solar.

"There's no green intent required here," CTO Colin Campbell said during a tour of the microgrid. "It's a good economic choice that also happens to be carbon-free."

The business model addresses a long-standing challenge in the energy storage sector. For over a decade, companies have been promising to build grid-scale storage from used EV batteries, but they've only materialized in small amounts. Redwood, which got its start as a battery materials and recycling company, is creating a new line that promises to deliver gigawatts of much needed energy storage in just a few years.

"This really demonstrates how economical the waste hierarchy actually is," Jessica Dunn, a battery expert at the Union of Concerned Scientists, told TechCrunch. That a large recycler like Redwood recognized the profit potential in repurposed EV batteries shows "where this end-of-life market will go," she added.

Repurposing batteries is a clear business opportunity for Redwood, but it might also be a business imperative. Redwood was founded to build a supply chain that could handle the predicted wave of used EV batteries that will hit the market. But that wave hasn't materialized quite as quickly as some predicted.

"If Redwood didn't enter the repurposing market, then they wouldn't get a share from the repurposed battery. They'd have to wait the five, 10, 15 years until they retired," she said. In the meantime, other companies would be able to sell the batteries for grid-scale storage, cutting Redwood out of years of revenue.

Straubel acknowledged this, noting in an interview that in many ways Redwood Materials started a bit early.

"We started really early, and in a way we started Redwood almost too early," he said, noting the company initially was collecting consumer batteries and production scrap ahead of the coming wave of EVs.

The current state of the recycling market underscores the challenge. "Right now, the recycling market is mostly manufacturing scrap, consumer electronics, and EV batteries that have failed under warranty," Dunn said. That has been enough for Redwood to process over 20 gigawatt-hours annually. But it pales in comparison to the 350 gigawatt-hours in EVs today and the 150 gigawatt-hours expected to hit the road every year.

Redwood currently has a recycling facility at its 175-acre campus in Sparks, Nevada, and it's developing a 600-acre facility in Charleston, South Carolina. The latter will remanufacture cathode and anode copper foil, both of which contain critical minerals that the U.S. would prefer stayed within its borders.

The company previously said that it will be capable of making 100 gigawatt-hours annually of cathode active material and anode foil by the end of this year. By the end of the decade, it expects production to hit 500 gigawatt-hours.

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